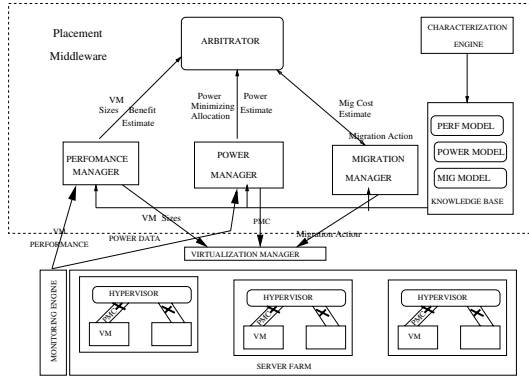




**Fig. 1.** *pMapper* Application Placement Architecture

middleware. In *pMapper*, all the power management actions are communicated by three different managers, with an arbitrator ensuring consistency between the three actions. The soft-actions like VM re-sizing and idling are communicated by the *Performance Manager*, that has a global view of the application in terms of QoS met and performance SLA. *Power Manager* triggers power management at a hardware layer whereas a *Migration Manager* interfaces with the *Virtualization Manager* to trigger consolidation through VM live migration.

The resource management flow of *pMapper* starts with the *Monitoring engine*, which collects the current performance and power characteristics of all the VMs and physical servers in the farm. *Performance Manager* looks at the current performance and recommends a set of target VM sizes based on the SLA goals. In case, the target VM sizes are different from the current VM sizes, it also presents an estimate of the benefit due to resizing. Similarly, *Power Manager* looks at the current power consumption and may suggest throttling (by DVFS or explicit CPU throttling). The central intelligence of *pMapper* lies in *Arbitrator*, which explores the configuration space for eligible VM sizes and placements and implements an algorithm to compute the best placement and VM sizes, based on the estimates received from *Performance*, *Power* and *Migration* managers.

*Performance Manager* supports interfaces using which the *Arbitrator* can query for the estimated benefit of a given VM sizing and placement (for all the VMs). In order to cater for heterogeneous platforms, the *Performance Manager* consults a *Knowledge Base* to determine the performance of an application, if one of its VM is migrated from one platform to another. Similarly, *Power Manager* supports interfaces using which *Arbitrator* can get the best power-minimizing placement for a given set of VM sizes. Also, *Power Manager* uses a power model in the *Knowledge Base* to determine the placement, as well as estimate the power for a given placement. *Migration Manager* estimates the cost of moving from a given placement to a new placement and uses the *Migration Model* for making the estimate. Once the *Arbitrator* decides on a new configuration, *Performance Manager*, *Power Manager*, and *Migration Manager* execute the VM sizing, server throttling and live migration operations, respectively.